

Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability

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The Gap in Optimization Theory

Classical View (The Descent Lemma)

- ▶ Gradient Descent (GD) is guaranteed to be stable **ONLY IF**:

$$\eta \leq \frac{2}{S(\theta)}, \quad \text{where } \eta \text{ is the learning rate.}$$

- ▶ $S(\theta) := \lambda_{\max}(\nabla^2 L(\theta))$ is the *Sharpness* of the loss landscape.

Deep Learning Reality

- ▶ Practitioners use large learning rates η that **blatantly violate** this bound.
- ▶ *Classical Expectation*: Exponential divergence (Blow-up).
- ▶ *Empirical Reality*: The network converges and minimizes the loss. **Why?**

Two Key Empirical Phenomena

To understand this mystery, we must look at two highly robust training phenomena:

1. Progressive Sharpening

- ▶ Early in training, sharpness $S(\theta)$ doesn't stay bounded.
- ▶ It steadily rises until it hits the critical instability threshold: $S(\theta) \approx 2/\eta$.

2. Edge of Stability (EOS)

- ▶ Once reaching $2/\eta$, the sharpness hovers persistently at this boundary.
- ▶ The loss exhibits short-term, high-frequency spikes but **decreases globally**.

Problem Statement

Our Focus

- ▶ Uncover the **implicit bias** of GD that prevents divergence at the EOS.
- ▶ Demonstrate how discrete GD implicitly tracks a *constrained optimization trajectory*.

Previous Work

- ▶ **2021: Phenomenon Discovered (Cohen et al.)**
 - ▶ First identified Progressive Sharpening EOS empirically.
- ▶ **2022: Origins Explained (Li et al. / Ahn et al.)**
 - ▶ Mathematically proved why networks naturally sharpen (scale-invariant properties).
- ▶ **2023: Stability Solved (Damian et al. Arora et al.)**
 - ▶ Damian et al. introduced cubic Taylor expansions and a self-stabilizing negative feedback loop.

GD and PGD

The Gradient Descent actually solves a Projected Gradient Descent implicitly. To see this, we first need to derive the formula of GD and PGD.

$$\textbf{(GD)} \quad \theta_{t+1} = \theta_t - \eta \nabla L(\theta_t), \quad (1)$$

$$\textbf{(PGD)} \quad \theta_{t+1}^\dagger = \text{proj}_M \left(\theta_t^\dagger - \eta \nabla L(\theta_t^\dagger) \right), \quad (2)$$

where

$$M = \left\{ S(\theta) \leq \frac{2}{\eta}, \nabla L(\theta) \cdot u(\theta) = 0 \right\},$$

$\theta \in \mathbb{R}^d$ is the parameter vector,

$L(\theta)$ is the empirical loss, $\nabla L(\theta)$ its gradient,

and $u(\theta)$ is the top eigenvector of the Hessian $\nabla^2 L(\theta)$.

Derivation of PGD

The author uses KKT method to derive simpler form of PGD:

$$\theta_{t+1}^{\dagger} = \text{proj}_M \left(\theta_t^{\dagger} - \eta \nabla L(\theta_t^{\dagger}) \right).$$

Equivalently, this projection can be written as the constrained optimization problem.s

Let $\tilde{\theta} = \theta_t^{\dagger} - \eta \nabla L(\theta_t^{\dagger})$, one can get

$$\begin{aligned} \theta_{t+1}^{\dagger} = \arg \min_{\theta} \quad & \frac{1}{2} \left\| \theta - \tilde{\theta} \right\|^2 \\ \text{s.t.} \quad & S(\theta) \leq \frac{2}{\eta}, \\ & \nabla L(\theta) \cdot u(\theta) = 0. \end{aligned} \tag{3}$$

KKT Conditions for PGD

Define the Lagrangian

$$\begin{aligned}\mathcal{L}(\theta, x_t, y_t) = & \frac{1}{2} \|\theta - \tilde{\theta}\|^2 + x_t \left(S(\theta) - \frac{2}{\eta} \right) \\ & + y_t (\nabla L(\theta) \cdot u(\theta)),\end{aligned}\tag{4}$$

where $\lambda \geq 0$ is the multiplier for the inequality constraint, and $\mu \in \mathbb{R}$ is the multiplier for the equality constraint.

The KKT conditions are

$$\theta - \tilde{\theta} + x_t \nabla S(\theta) + y_t \nabla_{\theta} (\nabla L(\theta) \cdot u(\theta)) = 0$$

$$\begin{aligned}S(\theta) - \frac{2}{\eta} & \leq 0, \\ \nabla L(\theta) \cdot u(\theta) & = 0, \\ x_t & \geq 0, \\ x_t \left(S(\theta) - \frac{2}{\eta} \right) & = 0,\end{aligned}\tag{5}$$

where $\nabla S(\theta)$ is the gradient of the sharpness $S(\theta)$.

Main Assumption

1. Progressive sharpening:
 $\alpha_t := -\nabla L \cdot \nabla S > 0$, the sharpness growth rate along GD.
2. Single unstable direction: $\lambda_2(\nabla^2 L) < 2/\eta$.
3. Bounded higher-order derivatives: $\nabla^3 L$ bounded and smooth.
4. Non-degenerate stabilization: $\alpha_t = \Theta(1)$, $\|\nabla S_t^\perp\| = \Theta(\rho_3)$.
5. Benign orthogonal directions: $\nabla^2 L$ and $\nabla^3 L$ behave regularly on $u_t^\perp$.

Use Lipschitz Property to achieve approximation

$$\theta_{t+1}^\dagger - \tilde{\theta} + x \nabla S(\theta_{t+1}^\dagger) + y \nabla_\theta (\nabla L(\theta) \cdot u(\theta))|_{\theta=\theta_{t+1}^\dagger} = 0$$

Use the Property that if learning step is not large,
 $\nabla S(\theta)$, $\nabla L(\theta)$, $u(\theta)$ these quantity are Lipschitz function(they would not change a lot).

$$\begin{aligned}\nabla S(\theta_{t+1}^\dagger) &= \nabla S(\theta_t^\dagger) + O(\eta \rho_3^2 \cdot \eta \|\nabla L_t\|) \\ &= \nabla S_t + O(\eta^2 \rho_3^2 \|\nabla L_t\|),\end{aligned}$$

$$\begin{aligned}\nabla_\theta (\nabla L(\theta) \cdot u(\theta))|_{\theta=\theta_{t+1}^\dagger} &= \nabla^2 L_{t+1} \cdot u_{t+1} + \nabla L_{t+1} \cdot \nabla u_{t+1} \\ &= S_{t+1} \cdot u_{t+1} + O(\eta \rho_3 \|\nabla L_{t+1}\|) \\ &= S_t \cdot u_t + O(\eta \rho_3 \|\nabla L_t\|),\end{aligned}$$

$$\begin{aligned}\|S_{t+1} u_{t+1} - S_t u_t\| &= \|S_{t+1}(u_{t+1} - u_t) - (S_{t+1} - S_t)u_t\| \\ &\leq \left|\frac{2}{\eta}\right| O(\eta \rho_3 \|\nabla L_t\|) + \rho_3 \|\theta_{t+1}^\dagger - \theta_t^\dagger\| \\ &= O(\rho_3 \eta \|\nabla L\|).\end{aligned}$$

Further use KKT to determine Lagrange Coefficients

Thus

$$\theta_{t+1}^{\dagger} = \theta_t^{\dagger} - \eta \nabla L_t - x_t S_t \boxed{u} - y_t \boxed{\nabla S_t} + O\left(\left(|x_t| \eta \rho_3 + |y_t| \eta^2 \rho_3^2\right) \|\nabla L_t\|\right).$$

First, we decompose $\nabla S = \langle \nabla S, u \rangle u + \nabla S^{\perp}$

$$\theta_{t+1}^{\dagger} = \theta_t^{\dagger} - \eta \nabla L_t - x' u_t - y' \frac{\nabla S_t^{\perp}}{\|\nabla S_t^{\perp}\|} + O(\epsilon^2 \cdot \eta \|\nabla L_t\|).$$

Then again use KKT, we can get further information of x' and y' .

Approximate Projected Gradient Dynamics

$$\theta_{t+1}^\dagger = \theta_t^\dagger - \eta \nabla L_t - x' u_t - y' \frac{\nabla S_t^\perp}{\|\nabla S_t^\perp\|} + O(\epsilon^2 \cdot \eta \|\nabla L_t\|) .$$

Applying KKT again gives additional information on the correction terms:

$$x' = O(\epsilon^2 \eta \|\nabla L_t\|) ,$$

$$y' = -\eta \left\langle \nabla L_t, \frac{\nabla S_t^\perp}{\|\nabla S_t^\perp\|} \right\rangle + O(\epsilon^2 \eta \|\nabla L_t\|) .$$

Substituting back yields

$$\theta_{t+1}^\dagger = \theta_t^\dagger - \eta P_{u_t, \nabla S_t}^\perp \nabla L_t + O(\epsilon^2 \eta \|\nabla L_t\|) .$$

Analyze the displacement $\theta_{t+1} - \theta_{t+1}^\dagger$

$$\left\{ \begin{array}{l} \theta_{t+1} = \theta_t - \eta \nabla L(\theta_t) \\ \quad = \theta_t - \eta \nabla L_t - \eta \nabla^2 L_t v_t - \frac{1}{2} \eta \nabla^3 L_t(v_t, v_t) + O(\eta \rho_4 \|v_t\|^3), \\ \theta_{t+1}^\dagger = \theta_t^\dagger - \eta P_{u_t, \nabla S_t}^\perp \nabla L_t + O(\epsilon^2 \eta \|\nabla L_t\|). \end{array} \right.$$

$$\begin{aligned} v_{t+1} &= (I - \eta \nabla^2 L_t) v_t - \eta \left[\frac{\nabla L \cdot \nabla S^\perp}{\|\nabla S^\perp\|^2} \right] \nabla S_t^\perp \\ &\quad - \frac{1}{2} \eta x_t^2 \nabla S_t - \eta x_t \nabla^3 L_t(u_t, v_t^\perp) \\ &\quad + O(\eta \rho_3 \epsilon \|v_t\|^2 + \eta \rho_4 \|v_t\|^3 + \epsilon^2 \eta \|\nabla L_t\|) \end{aligned} \tag{6}$$

Decompose displacement into two main direction

$$\begin{aligned} v_{t+1} = & (I - \eta \nabla^2 L_t) v_t \quad (\text{constant} + \text{quadratic}) \\ & - \eta \left[\frac{\nabla L \cdot \nabla S^\perp}{\|\nabla S^\perp\|^2} \right] \nabla S_t^\perp \quad (\text{first-order effect}) \\ & - \frac{1}{2} \eta x_t^2 \nabla S_t - \eta x_t \nabla^3 L_t(u_t, v_t^\perp) \quad (\text{third-order terms}) \\ & + O(\eta \rho_3 \epsilon \|v_t\|^2 + \eta \rho_4 \|v_t\|^3 + \epsilon^2 \eta \|\nabla L_t\|) \end{aligned}$$

It is noted that one of the main direction is u_t . Thus we want to decompose the displacement into $P_{u_t} + P_{u_t}^\perp$ two parts. And in the orthogonal part, $\nabla S^\perp$ is one of a dominant term.

Decompose displacement into two main direction

Define $x_t = v_t \cdot u_t$, and $y_t = v_t \cdot \nabla S_t^\perp$ one can get

$$x_{t+1} = -x_t(1 + \eta y_t) - \frac{1}{2}\eta x_t^2 \nabla S_t \cdot u_t + O(E_t),$$

$$v_{t+1}^\perp = P_{u_{t+1}}^\perp \left[(I - \eta \nabla^2 L_t) P_{u_t}^\perp v_t + \eta \nabla S_t^\perp \left(\frac{\delta_t^2 - x_t^2}{2} \right) \right] + O(E_t),$$

where

$$E_t := \epsilon^2 \delta \max \left(1, \left(\frac{\|v_t\|}{\delta} \right)^3 \right),$$

and

$$\delta_t = \sqrt{\frac{2\alpha_t}{\beta_t}}, \quad \beta_t = \|\nabla S_t^\perp\|^2,$$

where β_t is the stabilization strength, and δ_t controls the deviation size.

Predicted Dynamic

Therefore, we define the predicted dynamics by dropping the error terms:

$$x_{t+1}^* = -x_t^*(1 + \eta y_t^*),$$

$$(v_{t+1}^*)^\perp = P_{u_{t+1}}^\perp \left[(I - \eta \nabla^2 L_t) P_{u_t}^\perp v_t^* + \eta \nabla S_t^\perp \left(\frac{\delta_t^2 - (x_t^*)^2}{2} \right) \right].$$

Lemma (2-Step Approximation)

Let

$$r_t := v_{t+2} - v_{t+2}^*,$$

where v_{t+2}^* is generated by the predicted dynamics.

Assume

$$\|v_t\| \leq \epsilon^{-1} \delta.$$

Then

$$\|r_t\| \leq O\left(\epsilon^2 \delta \cdot \max\left(1, \left(\frac{\|v_t\|}{\delta}\right)^3\right)\right).$$

Predicted Dynamics

Actually, define $A_t = (I - \eta \nabla^2 L_t) P_{u_t}^\perp$, which is the effect of second order term of orthogonal part, one can get

$$\begin{aligned} (v_{t+1}^*)^\perp &= P_{u_{t+1}}^\perp \left[A_t v_t^* + \eta \nabla S_t^\perp \left(\frac{\delta_t^2 - (x_t^*)^2}{2} \right) \right] \\ &= \eta \sum_{s=0}^t P_{u_{t+1}}^\perp \left[\prod_{k=t}^{s+1} A_k \right] \nabla S_s^\perp \left(\frac{\delta_s^2 - (x_s^*)^2}{2} \right). \end{aligned}$$

Thus

$$y_{t+1}^* = \eta \sum_{s=0}^t \beta_{s \rightarrow t} \left(\frac{\delta_s^2 - (x_s^*)^2}{2} \right), \quad (7)$$

where

$$\beta_{s \rightarrow t} := (\nabla S_{t+1}^\perp)^\top \left[\prod_{k=t}^{s+1} A_k \right] \nabla S_s^\perp, \quad (8)$$

which generalizes β by measuring the change in sharpness between steps s and $t + 1$.

Simplified Dynamics

If we further perform some approximation. If we omit the effect of second order term, then

$$x_{t+1} = -(1 + \eta y_t)x_t,$$

$$y_{t+1} = y_t + \eta \left(\alpha - \frac{\beta}{2} x_t^2 \right),$$

where

$\beta_{s \rightarrow t}$ collapses to the constant $\beta = \|\nabla S\|^2$ and $\alpha = -\nabla L \cdot \nabla S$.

Key Result:

Theorem

Let

$$\mathcal{T} := O(\epsilon^{-1})$$

and assume that

$$\min_{t \leq \mathcal{T}} |x_t^*| \geq c_1 \delta.$$

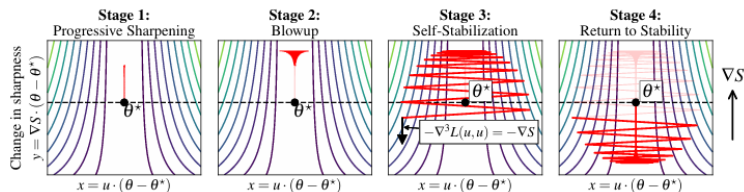
Then for any $t \leq \mathcal{T}$, we have

$$\theta_t = \theta_t^\dagger + v_t^* + O(\epsilon \delta), \quad (\text{Deviation from } \theta^\dagger), \quad (9)$$

$$L(\theta_t) = L(\theta_t^\dagger) + \frac{(x_t^*)^2}{\eta} + O\left(\frac{\epsilon \delta^2}{\eta}\right), \quad (\text{Loss}) \quad (10)$$

$$S(\theta_t) = \frac{2}{\eta} + y_t^* + (\nabla S_t \cdot u_t) x_t^* + O\left(\frac{\epsilon^2}{\eta}\right). \quad (\text{Sharpness}) \quad (11)$$

Analysis of Dynamics



The EOS behavior can be understood as a negative feedback loop.

$$x_{t+1} = -(1 + \eta y_t)x_t, \quad y_{t+1} = y_t + \eta \left(\alpha - \frac{\beta}{2} x_t^2 \right)$$

- ▶ **Stage 1: Progressive sharpening.** When x_t is small, $\alpha > 0$ dominates, so sharpness increases.
- ▶ **Stage 2: Blowup.** Once $y_t > 0$, the unstable displacement $|x_t|$ grows.
- ▶ **Stage 3: Self-stabilization.** Large $|x_t|$ activates the cubic term, pushing the dynamics toward $-\nabla S$.
- ▶ **Stage 4: Return to stability.** Sharpness falls below $2/\eta$, so $|x_t|$ shrinks and the cycle repeats.

Toy Reproduction

In the Appendix-B of the original paper, we reproduce the mechanism with the toy loss that:

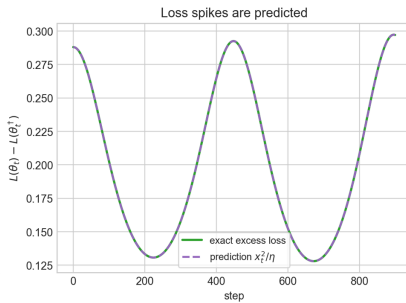
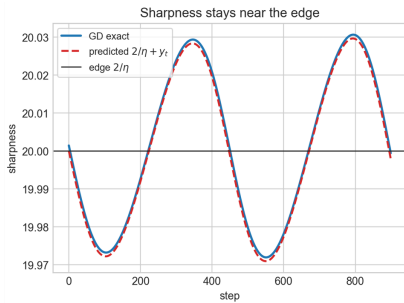
$$L(x, y, z) = \frac{1}{2} \left(\frac{2}{\eta} + \sqrt{\beta} y \right) x^2 - \frac{\alpha}{\sqrt{\beta}} y + z$$

Writing $Y_t = \sqrt{\beta} y_t$, GD exactly reduces to

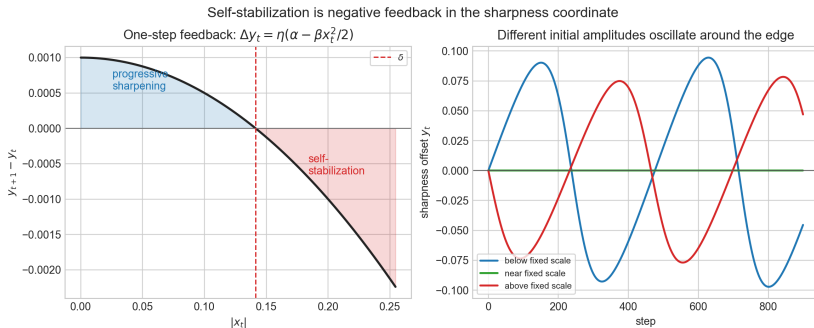
$$x_{t+1} = -(1 + \eta y_t) x_t, \quad y_{t+1} = y_t + \eta \left(\alpha - \frac{\beta x_t^2}{2} \right)$$

- ▶ Simulation-only reproduction; no real-world dataset is used.
- ▶ Parameters: $\eta = 0.1$, $\alpha = 0.01$, $\beta = 1$, $x_0 = 1.2\delta$.
- ▶ Fixed scale: $\delta = \sqrt{2\alpha/\beta}$.

Toy Reproduction

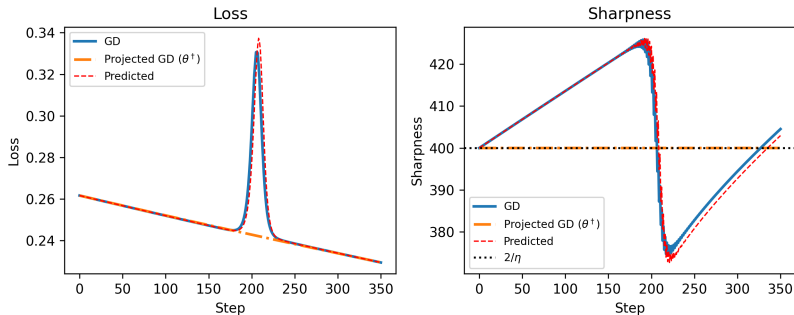


Self-Stabilization Is Negative Feedback



- ▶ Below δ , the model sharpens.
- ▶ Above δ , the sharpness correction turns negative.
- ▶ This is the mechanism behind the coupling theorem.

Real Data Experiment(CIFAR10, 3-layer MLP, $\eta = 0.005$)



We plot the learning trajectory of:

Gradient Descent $\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$

Projected Gradient $\theta_{t+1}^\dagger = \text{proj}_M(\theta_t^\dagger - \eta \nabla L(\theta_t^\dagger))$

Predicted dynamics $L(\theta_t) = L(\theta_t^\dagger) + \frac{(x_t^*)^2}{\eta}$

Real Data Experiment(CIFAR10, 3-layer MLP, $\eta = 0.005$)

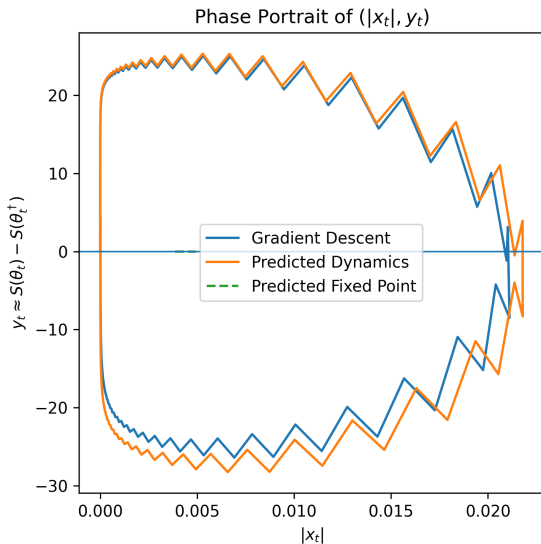


Figure 1: Phase Diagram

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